ICSR CASE PROCESSING STEPS

Professional & Systematic Pharmacovigilance Notes

Video	“ICSR Case Steps in Pharmacovigilance: Full Tutorial”
Focus	End-to-end processing of an Individual Case Safety Report (ICSR)
Use	Pharmacovigilance study, internship training, interviews and practical case-processing revision

These are structured study notes based on the requested video topic and standard ICSR/pharmacovigilance principles. Exact workflows, timelines, coding conventions and database fields vary by organization and jurisdiction; follow the applicable controlled SOP for real case processing.

1. What is an ICSR?

An Individual Case Safety Report (ICSR) is a report containing information describing a suspected adverse reaction/event associated with one patient and one or more medicinal products. ICSRs are a fundamental unit of individual case safety information in pharmacovigilance.

- ICSRs can originate from healthcare professionals, patients/consumers, literature, clinical studies, regulators, patient-support programs and other valid sources.
- The purpose of processing is to transform source information into a structured, medically meaningful and traceable safety case.
- A case may be updated through follow-up reports as new information becomes available.

2. Overall ICSR Case-Processing Workflow

Case receipt / intake

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Minimum criteria & case validity

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Case creation & identification

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Duplicate search

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Data entry / data abstraction

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Medical & product coding

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Seriousness / causality / expectedness assessment

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Narrative preparation

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Medical review / quality control

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Regulatory reporting or database finalization

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Follow-up and case closure / subsequent updates

The order can differ between organizations; some activities occur in parallel and some require medical-review approval.

3. Step 1 — Case Receipt / Intake

- Record when and how the report was received.
- Identify source type and source document/reference.
- Preserve the original report and attachments according to document-retention procedures.
- Check whether the report is a new case or information relating to an existing case.
- Protect personal and confidential information throughout intake.

4. Step 2 — Determine Whether the Report is a Valid ICSR

A report generally needs enough information to identify a patient, a reporter, a suspected medicinal product and a suspected adverse event/reaction. The exact minimum criteria depend on the applicable regulatory framework and SOP.

- Identifiable patient: sufficient information to distinguish the patient.
- Identifiable reporter: sufficient information to identify the source/reporter.
- Suspect product: medicinal product suspected in relation to the event.
- Adverse event/reaction: medical occurrence reported in association with the product.
- If essential information is missing, determine whether the report can be considered valid or requires clarification/follow-up.

Do not create validity by guessing missing information. Document the gap and follow the applicable procedure.

5. Step 3 — Create the Case and Assign an Identifier

- Assign internal case number according to safety-system convention.
- Record initial/follow-up status where applicable.
- Capture receipt date and source country as required.
- Link the source document and previous case number when the report is a follow-up.
- Maintain traceability between source report and database record.

6. Step 4 — Duplicate Search

- Compare patient characteristics and demographics.
- Compare suspect products, dose and exposure dates.
- Compare event terms, onset date and clinical course.
- Compare reporter/source details.
- Compare narrative and supporting documents.
- If a potential duplicate is found, follow the organization's duplicate-management process.

7. Step 5 — Data Abstraction and Entry

Data abstraction means extracting relevant information from the source report and entering it accurately into the safety database or controlled classification sheet.

- Patient: age/date of birth, sex, weight and relevant history.
- Event: verbatim description, onset, duration, outcome, investigations and treatment.
- Suspect product: name, active ingredient, dose, route, frequency, indication and therapy dates.
- Concomitant products: other relevant medicines/products and dates.
- Reporter/source: profession, country and contact/reference information as permitted.
- Follow-up: missing clinically important information and questions needed.

8. Step 6 — Verbatim Information vs Coded Information

Type	Meaning	Example
Verbatim	Information exactly or substantially as reported	“Red itchy rash all over body”
Coded	Standardized terminology representing the clinical concept	Appropriate MedDRA term selected from the current version
Narrative	Chronological clinical summary of the case	Exposure → onset → management → outcome → assessment

Key principle: preserve what was reported before applying interpretation or coding.

9. Step 7 — MedDRA Coding

MedDRA is the standardized medical terminology widely used for coding adverse events/reactions and other medical concepts in regulated safety information.

- Read the full clinical description before selecting a term.
- Preserve reporter wording in the verbatim field.
- Select the most appropriate current MedDRA term supported by the documented concept.
- Do not introduce an unsupported diagnosis.
- Apply the current MedDRA version and organization-specific conventions.
- Use medical review/clarification when the source does not support a definitive concept.

10. Step 8 — Product / Drug Coding

- Standardize medicinal product name and active ingredient.
- Capture strength, formulation and route when relevant.
- Use controlled product dictionaries or validated coding tools where available.
- Do not assume manufacturer, batch or formulation if not documented.
- Maintain the relationship between suspect product and reported indication/event.

11. Step 9 — Seriousness Assessment

Seriousness is a regulatory classification, not simply the intensity of an event. Common seriousness criteria include:

- Death
- Life-threatening event
- Initial or prolonged hospitalization
- Persistent or significant disability/incapacity
- Congenital anomaly/birth defect
- Other medically important event

An event may be severe but non-serious, or serious because it meets a regulatory criterion. Do not use “severe” and “serious” interchangeably.

12. Step 10 — Outcome Assessment

- Recovered/resolved
- Recovering/resolving
- Not recovered/not resolved

- Recovered with sequelae
- Fatal
- Unknown

Use the outcome supported by the latest available information. Update the case when follow-up changes the outcome.

13. Step 11 — Causality Assessment

Causality assessment estimates the likelihood that the medicinal product is related to the reported event. PvPI uses the WHO–UMC causality assessment framework for ADR causality assessment.

- Review temporal relationship.
- Consider alternative causes including disease, concomitant medicines and other exposures.
- Review dechallenge where clinically available.
- Review rechallenge only if it actually occurred; do not intentionally rechallenge solely to establish causality when unsafe/unethical.
- Consider pharmacological and clinical plausibility.
- Document the conclusion according to the required framework/SOP.

14. WHO–UMC Categories — Quick Review

Category	Core idea
Certain	Strong relationship; plausible timing and no reasonable alternative explanation, with supportive dechallenge/rechallenge evidence where relevant.
Probable/Likely	Reasonable timing and alternatives unlikely; clinically reasonable response to withdrawal; rechallenge not required.
Possible	Reasonable timing but disease or another drug could also explain the event.
Unlikely	Timing makes a relationship improbable and another explanation is plausible.
Conditional/Unclassified	More information is required or additional information is being examined.
Unassessable/Unclassifiable	Information is insufficient or contradictory and cannot be supplemented/verified.

15. Step 12 — Expectedness / Listedness

Expectedness asks whether the nature and specificity of the reported reaction are consistent with the applicable reference safety information. The exact reference document depends on the product and reporting context.

- Use the correct current reference safety information.
- Do not infer expectedness merely because the medicine has other known adverse effects.
- Document the reference and assessment where required.
- Expectedness is separate from seriousness and causality.

16. Step 13 — Special Situations

Depending on product and SOP, identify special situations such as:

- Pregnancy/lactation exposure
- Medication error
- Overdose
- Off-label use
- Misuse
- Abuse
- Occupational exposure
- Lack of efficacy
- Product quality complaint
- Drug interaction
- Special population exposure

A special situation may occur with or without an adverse event. Apply reporting rules relevant to the case.

17. Step 14 — Write the ICSR Narrative

The narrative is a concise chronological clinical summary that allows a reviewer to understand the case without repeatedly referring to the source document.

- Patient background relevant to the case.
- Suspect product: name, indication, dose, route and therapy dates.
- Sequence and timing of the event.
- Clinical signs/symptoms and relevant investigations.
- Management and action taken with the suspect product.
- Dechallenge/rechallenge information when available.
- Outcome.
- Relevant concomitant medicines and alternative causes.
- Causality/medical assessment where required.

Narrative principle: chronological, factual, concise, medically meaningful and traceable.

18. Example Narrative Structure

Example: A 45-year-old female received Drug X 500 mg orally twice daily for a bacterial infection from 1 October 2026. On 3 October 2026, she developed a generalized maculopapular rash. Drug X was discontinued and supportive treatment was provided. The rash subsequently improved. No other new medicinal products were reported. Based on the available information, the case was assessed according to the applicable causality framework and classified as [category] by the responsible reviewer.

Fictional training example. The causality category should only be assigned after reviewing complete case information.

19. Step 15 — Medical Review

- Review clinical accuracy and event interpretation.
- Confirm seriousness and outcome.
- Review causality assessment.
- Review MedDRA coding where medical judgment is required.
- Assess whether narrative accurately represents source information.
- Identify follow-up questions and medically important missing data.
- Approve or return the case according to workflow.

20. Step 16 — Quality Control (QC)

- Patient and reporter information checked.
- Suspect product correctly identified.
- Event terms and dates consistent.
- Seriousness criteria supported by source.
- Outcome consistent with latest information.
- Causality/expectedness documented as required.
- MedDRA/product coding reviewed.
- Narrative matches database fields and source.
- Duplicate status assessed.
- Follow-up needs identified.
- Case status and reviewer details complete.

21. Step 17 — Regulatory Reporting / Case Finalization

After required processing and review, the case may proceed to regulatory reporting or finalization according to applicable reporting rules, timelines and jurisdiction.

- Confirm whether expedited reporting criteria apply.
- Confirm required data elements and reporting format.
- Use validated transmission mechanisms.
- Document submission/status information.
- Maintain audit trail and source documentation.
- Update the case when valid follow-up information is received.

22. Follow-up Process

- Identify missing clinically significant information.
- Send focused questions to reporter/source.
- Record follow-up date and status.
- Update the case with newly received information.
- Reassess seriousness, outcome, causality or expectedness if new information changes the assessment.
- Maintain previous information and traceability according to database/audit-trail rules.

23. Common ICSR Processing Errors

- Creating a case without confirming minimum validity criteria.
- Failing to search for duplicates.
- Changing reporter wording into a diagnosis without adequate support.
- Confusing seriousness with severity.
- Selecting causality solely from temporal association.
- Using outdated MedDRA terminology.
- Ignoring concomitant medicines or alternative causes.
- Writing a narrative that contradicts database fields.
- Failing to update outcome after follow-up.
- Missing required regulatory timelines.
- Storing identifiable information outside controlled systems.

24. Practical ICSR Processing Checklist

- ☐ Source received and documented
- ☐ Validity criteria assessed
- ☐ Case ID assigned
- ☐ Duplicate search completed
- ☐ Patient data entered
- ☐ Suspect product entered/coded
- ☐ Event entered/coded
- ☐ Concomitant products assessed
- ☐ Seriousness assessed
- ☐ Outcome assessed
- ☐ Causality assessed

- ☐ Expectedness/listedness assessed
- ☐ Special situations checked
- ☐ Narrative prepared
- ☐ Medical review completed
- ☐ QC completed
- ☐ Reporting/finalization status documented
- ☐ Follow-up plan recorded

25. ICSR Case Processing — One-Page Revision Flow

SOURCE REPORT

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VALIDATE ICSR

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CASE CREATION

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DUPLICATE CHECK

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DATA ENTRY

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MEDDRA + PRODUCT CODING

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SERIOUSNESS + OUTCOME

↓

CAUSALITY + EXPECTEDNESS

↓

SPECIAL SITUATIONS

↓

NARRATIVE

↓

MEDICAL REVIEW

↓

QC

↓

REGULATORY REPORTING / FINALIZATION

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FOLLOW-UP & CASE UPDATE

26. Exam & Interview Questions

Q. What is an ICSR?

A. An individual case safety report describing a suspected adverse event/reaction associated with one or more medicinal products in an individual patient.

Q. What are the key minimum elements of a valid safety report?

A. An identifiable patient, identifiable reporter, suspected medicinal product and suspected adverse event/reaction, subject to the applicable reporting framework.

Q. Why is duplicate checking important?

A. To avoid processing the same safety information as separate cases.

Q. What is the difference between verbatim and coded terms?

A. Verbatim preserves the reported wording; coded terms standardize the clinical concept using controlled terminology.

Q. What is MedDRA?

A. A standardized medical terminology used for coding medical concepts in regulated safety information.

Q. Is seriousness the same as severity?

A. No. Seriousness uses defined regulatory criteria; severity describes intensity.

Q. What is the role of causality assessment?

A. To estimate the likelihood of a relationship between the suspect product and event using the required framework.

Q. What is an ICSR narrative?

A. A concise chronological clinical summary integrating relevant patient, product, event, management, outcome and assessment information.

Q. Why is medical review important?

A. It verifies clinical interpretation, coding, seriousness, causality, narrative quality and medically important follow-up needs.

27. Key Takeaways

- ICSR processing is a controlled, stepwise pharmacovigilance workflow.
- First establish validity, then create and trace the case.
- Duplicate detection prevents duplicate case processing.
- Accurate data abstraction preserves source information.
- Coding standardizes clinical and product information.
- Seriousness, outcome, causality and expectedness are separate assessments.
- The narrative should tell the clinical story chronologically and accurately.
- Medical review and QC are essential before finalization/reporting.
- Follow-up is part of the lifecycle of an ICSR.
- Always follow current company SOPs and applicable regulatory requirements.

28. References

1. ICH E2A — Clinical Safety Data Management: Definitions and Standards for Expedited Reporting.

2. ICH E2B(R3) — Individual Case Safety Reports (ICSRs): data elements and standards.

3. WHO / Uppsala Monitoring Centre — WHO–UMC standardized case causality assessment.

4. Indian Pharmacopoeia Commission (IPC), Pharmacovigilance Programme of India (PvPI) — pharmacovigilance and ADR reporting resources.

5. MedDRA MSSO — current MedDRA terminology and Introductory Guide.

Video reference: “ICSR Case Steps in Pharmacovigilance: Full Tutorial” — https://youtu.be/ryof8pJbX1o

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